

A Parseval Corridor for Direct Additive Twists on the Determinant-Two Möbius Core

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Abstract

The periodic-approximation route to an additive phase has a sharp minor-arc obstruction. We open a different, direct route by treating the phase as an L^2 variable. On the literal determinant-two two-Möbius core, Parseval gives an exact mean-square identity for the normalized additive twist, at every scale and without a scale-exceptional set. The same identity holds on every complete Fourier grid at least as long as the fiber interval. In the small-polylogarithmic envelope this is a positive fixed- X power in phase L^2 . It is not a pointwise theorem for a specified production phase, and therefore does not close the original pointwise additive-twist frontier.

Keywords. Möbius correlation; additive twist; Parseval identity; phase average; determinant two.

Claim boundary. The power saving below is in Lebesgue L^2 over the additive phase. It is not a bound at a distinguished phase and is not a production phase-cell registry. Parseval uses orthogonality rather than a new pointwise Möbius estimate. No physical H3 return, deterministic all-prefix theorem, $1/400$, prime-pair lower bound, or twin-prime theorem is proved.

1 The literal core and its phase transform

Put $e(x) = \exp(2\pi ix)$. Retain the determinant-two data

$$\begin{aligned} a, s \geq 1, \quad d, u \in \mathbb{Z}, \quad (a, s) = 1, \quad as \text{ odd}, \quad su - ad = 2, \\ q = as, \quad t(z) = ad + qz, \quad c_z = \mu(d + sz)\mu(u + az). \end{aligned}$$

For $N > 0$, let

$$I_N = \{z \in \mathbb{Z} : N < t(z) \leq 2N\}, \quad L_N = |I_N|, \quad E_N = \sum_{z \in I_N} |c_z|^2.$$

The interval I_N consists of consecutive integers, and

$$E_N \leq L_N \leq \frac{N}{q} + 1. \tag{1}$$

Define the normalized direct additive twist

$$F_N(\alpha) = \frac{q}{N} \sum_{z \in I_N} c_z e(-\alpha z), \quad \alpha \in \mathbb{T} = \mathbb{R}/\mathbb{Z}. \tag{2}$$

TPC-158 controls explicit major arcs through periodic approximation and proves that this approximation route is expensive on a uniform minor-arc family [1]. The definition (2) does not approximate the phase.

2 Exact phase mean square

Theorem 2.1 (Direct additive-twist Parseval identity). *For every $N > 0$,*

$$\boxed{\int_{\mathbb{T}} |F_N(\alpha)|^2 d\alpha = \frac{q^2}{N^2} E_N.} \quad (3)$$

In particular,

$$\|F_N\|_{L^2(\mathbb{T})}^2 \leq \frac{q}{N} + \frac{q^2}{N^2}. \quad (4)$$

No exceptional set of scales is required.

Proof. Expanding the square and using character orthogonality gives

$$\int_{\mathbb{T}} e(-\alpha(z - z')) d\alpha = \mathbf{1}_{\{z=z'\}}.$$

Thus every off-diagonal term vanishes and the diagonal equals $q^2 E_N / N^2$. Apply (1). $\square$

Corollary 2.2 (Exceptional phases). *For every $\lambda > 0$,*

$$\text{meas}\{\alpha \in \mathbb{T} : |F_N(\alpha)| > \lambda\} \leq \frac{q^2 E_N}{N^2 \lambda^2} \leq \frac{q/N + q^2/N^2}{\lambda^2}. \quad (5)$$

Proof. This is Chebyshev's inequality applied to (3). $\square$

The identity is stronger than a logarithmic estimate in its own declared norm, but weaker than a bound at any named α . It holds equally for arbitrary bounded coefficients; the Möbius form identifies the actual core rather than supplying the orthogonality.

3 A finite exact Fourier registry

Write $I_N = \{z_0, \dots, z_0 + L_N - 1\}$.

Proposition 3.1 (Complete-grid identity). *If $M \geq L_N$, then*

$$\frac{1}{M} \sum_{r=0}^{M-1} \left| F_N \left(\frac{r}{M} \right) \right|^2 = \frac{q^2}{N^2} E_N. \quad (6)$$

Consequently the number of grid phases for which $|F_N(r/M)| > \lambda$ is at most

$$\frac{M q^2 E_N}{N^2 \lambda^2}.$$

Proof. After expansion, the finite character average is one precisely when $M \mid z - z'$. Since $|z - z'| \leq L_N - 1 < M$, this occurs precisely on the diagonal. The counting assertion follows by summing $|F_N|^2 > \lambda^2$ over bad grid points. $\square$

The condition $M \geq L_N$ is structural. For a shorter grid, distinct exponents may collide modulo M , so (6) need not hold.

4 Interpretation of the power envelope

Suppose the small-polylogarithmic envelope

$$q \leq (\log X)^{\eta_0}, \quad N \geq \sqrt{X}$$

holds. For sufficiently large X , $q \leq N$, and (4) gives

$$\|F_N\|_{L^2(\mathbb{T})} \leq \sqrt{2} X^{-1/4} (\log X)^{\eta_0/2}. \quad (7)$$

Thus TPC-167 has the precise program status

PROVED_L1_ACTUAL_CORE_PHASE_METRIC_SINGLE_CELL.

Its analytic norm is L2_PHASE, while `program_positive_L2 = false` and `fixed_atom = false`. The positive exponent in (7) belongs to a phase-averaged norm. It cannot be entered as a production fixed-phase or endpoint-V3 payment.

5 Why pointwise selection remains open

The implication

$$\|F_N\|_{L^2(\mathbb{T})} \ll N^{-1/2} q^{1/2} \implies |F_N(\alpha_\star)| = o(1)$$

is false without information about the selected phase $\alpha_\star$. At the coefficient level, take $c_z = 1$ on an interval and $\alpha_\star = 0$. Then

$$|F_N(0)| = \frac{qL_N}{N},$$

which is of constant order when $L_N \sim N/q$, while Parseval still has mean square of order q/N . This example is a logical nonimplication for coefficient-blind phase averaging; it is not a claim that the literal Möbius coefficients realize the example.

Accordingly, the parent-ready direct-twist node is advanced only to a typed child:

direct additive twist $\longrightarrow$ phase- L^2 direct twist.

The original pointwise node still needs a source-locked phase registry plus an avoidance or selection theorem, or a genuinely pointwise arithmetic estimate.

6 Reproducible certificate

The standard-library audit source-locks TPC-158, checks Parseval on a shifted signed/zero fixture using two complete Fourier grids of different sizes, and verifies the normalization bound. These are finite implementation checks; the proof above establishes the continuous identity. Its executable mutation contract rejects promotion from phase L^2 to a fixed phase, from a phase power to endpoint V3, or from orthogonality to Möbius-specific cancellation. Hashes certify artifact integrity only.

7 Conclusion

The minor-arc obstruction to small-period approximation does not stop every direct route. Parseval gives an exact, exceptional-scale-free direct additive-twist theorem on the actual core, with a positive fixed- X power in phase L^2 . The cost of this advance is equally exact: a distinguished physical phase remains outside the theorem. TPC-168 asks how much of this phase-average information survives on a finite source-locked phase registry.

References

- [1] Liang Wang. An Exact Major-Arc Gate for Additive Phases: Periodic Projection and a Minor-Arc Route Obstruction, 2026. TPC-158 manuscript.